

UNIVERSITY OF ILLINOIS AT CHICAGO

Department of Chemical Engineering

CHE494: Entrepreneurship in Chemical Engineering

Instructor: Dr. Said Al-Hallaj

Assignment 3

Submitted By: Ian Lim

November 21, 2019

The Current War

This movie is a recount of 3 great inventors - Thomas Edison, George Westinghouse, and Nikola Tesla and their journey that they took to light up the World's Columbian Exposition in Chicago in 1893. The three already had significant contributions to electricity, and this paper seeks to analyze each character's traits, and discuss the intellectual property (IP) rights of each person.

Thomas Edison: By the 1890s, Edison was already celebrated for his multitude of inventions. Apart from the incandescent lamp, he was also attributed for inventing the phonograph and motion pictures. He was celebrated for being one of America's greatest inventors. But, his stance on electricity delivery to the town was reliant on Direct Current (DC). While it was allegedly safer, DC was not able to reach long distances (approximately 1 mile radius).

At the same time, Edison knew about the Alternating Current (AC) technology that George Westinghouse was developing. Feeling threatened by that, Edison filed a patent to protect his intellectual property and attempted to sue Westinghouse. Being an inventor, it is important to safeguard one's creations. Edison also knew that he wanted every single one - from the rich to the commoner, to be able to have access to electricity.

However, this came at a cost. While working hard to prove to the world that DC was the way to go, he neglected his personal life. He did not make it back in time when his wife passed away, and that was the sacrifice he had to make. His approach to solving a problem was "Everyone needs electricity, so let's provide them with that." He identified a problem, and this was where the entrepreneurial side of Edison showed up, albeit to a lesser extent than his inventor side. In my opinion, entrepreneurs should be humble and listen to constructive advice from their team. Edison had a great team, with bright and smart people to support him. Unfortunately, he was too prideful and did not value the advice of his team mates and that was what eventually led him to lose the current war.

George Westinghouse: Westinghouse was also an inventor, having discovered the AC technology which he argued was much cheaper, and could reach a longer range to more cities on one powerhouse. However, it was allegedly more dangerous, having enough energy to shock a horse to death as demonstrated by Edison. Westinghouse's approach was definitely more from the entrepreneurial side, since he felt a burning passion with the problem that he identified, which was the pain in the power grid.

Westinghouse's tenacity led him to deliver a short but powerful speech at his pitch for the Chicago world fair which eventually landed him the victory of the current war. At the end of the day, cost triumphs over all. Businesses are set up to make money, and Westinghouse's approach provided a higher return on investment. When Edison and Tesla's partnership failed, Westinghouse saw Tesla's potential and approached him to work on the motor that could power the electrical demand for the world fair. In return, Westinghouse offered Tesla \$2.50 for every volt that his motor could generate.

Being adaptive to the situations around him, and despite getting sued by Edison for copying his invention, Westinghouse was able to get around it because his technology relied on AC and not DC. In this scenario, it is important to note what one is protecting and the bounds of it. Loopholes like Westinghouse thus fall outside of the invention's boundary, thereby allowing Westinghouse to succeed as a result.

Nikola Tesla: Tesla was a smart and bright inventor. While he did not invent the lightbulb, he was able to come up with a motor that could supply enough power to the world fair. He was insistent on his idea, and eventually got Westinghouse to adopt it. With his smart intellect, however, he was not an entrepreneur due to his lack of social skills and eccentric behavior. However, one can agree that he definitely was an inventor. And this was why Westinghouse succeeded in the current war. Westinghouse was strong with his entrepreneurial background and Tesla had the intellect; with their strong collaboration, they were able to combine their strengths and bring the Westinghouse company to victory.